INSTRUCTOR LESSON PLAN

Where Does It Live?

Builds: Digital & Systems Fluency

A turnkey unit for teaching students to stop treating software as magic and start asking where their stuff goes and who can see it.

> **Builds:** Digital & Systems Fluency **Grade band:** 6–12 (adaptation notes for grades 3–5 and college/CTE below) **Time:** 3 sessions of ~45 minutes, run across one to three weeks (not three days in a row) **Format:** Guided whole-class audit → blank-template audit → fully student-designed audit

Why this unit

Here's the line from the keynote that stings: our students are excellent *users* of technology and terrible *understanders* of it. Ask a student what's inside a cell phone, or to organize a folder structure for their six or seven classes, and watch the confidence drain out of the room. They can open the app; they cannot say where the file went. That used to be a "nice to have." It is not anymore. The student who doesn't know where a file lives on their own machine is now the *least* equipped person in the room for the single tool redefining their future — because every interaction with AI is a question of *what did I just hand over, and where did it go?* AI moved this cliff from "nice to have" to the floor. This unit builds the floor.

Learning objective

By the end of the third audit, **students can independently trace where their own digital work lives, describe in plain terms what a tool does with what they give it, and ask "where does this go, and who can see it?" without being prompted** — the question has become a habit, not an assignment.

Materials & prep

Teacher prep (~15 min the first time, ~5 min after):

- Pick **three** of your own real school projects/workflows for the live demo so you can model honestly ("Here's where *my* gradebook export actually lives — let's find out together"). It's fine to not know the answer; finding out *is* the lesson.
- Print the **Student Handout** (below), one per student, or post it to your LMS. Print a few extra blank copies of just the diagram box for Session 2.
- Make sure students can reach the devices and accounts they actually use — their school Google/Microsoft account, the LMS, any AI tool you permit. Nothing here requires admin access; it's all visible from a normal student login.

Students need: the handout, their own device logged into their own school accounts, and something to write/draw with (a blank diagram drawn by hand is the point — see the keynote).

The arc at a glance

Session	Focus	~Time
1 — Guided audit	Map where three real projects live; trace what "submit" actually does, with a guided worksheet + whole-class debrief	45 min

Session	Focus	~Time
2 — Blank template	Audit a *different* tool/workflow using only a blank diagram — no prompts	45 min
3 — Self-directed	Pick the tool, design the diagram, write your own questions; "what will I do differently?" close	45 min

Step-by-step sequence

Session 1 — Guided audit (45 min)

- **(5 min) Frame the question.** Tell students plainly: *"Software is not magic. Everything you make lives somewhere, and someone — or something — can see it. Today we find out where your stuff actually is."* Ask the keynote provocation out loud: "Can anyone tell me, right now, where your last assignment is saved? Not the app — the actual place." Let the uncertainty sit.
- **(15 min) Map three projects (guided worksheet).** Walk the class through the handout's file-audit table for three of their own recent projects. For each: What's it called? What *folder* is it in? Is it on this device, in the cloud, or both? *Which account* — your school login, a personal one, an app's own storage? Do the first one together on the board with your own example; students do the next two.
- **(15 min) Trace the "submit."** Students pick one assignment they turned in and draw, on the handout, the path it took when they hit submit: my device → ... → where it lands → who can open it. Most will discover they have no idea what happens in the middle. Good. That gap is the point.
- **(10 min) Whole-class debrief.** Surface what surprised them. Common jaw-droppers: "I have the same file in three places and don't know which is the real one," "I don't know if that's my school account or my personal one," "I never knew the teacher could see *when* I submitted." Name the habit you're building: *every time you use a tool, ask where it lives and who can see it.*

Session 2 — Blank template, different tool (45 min)

- **(5 min) Raise the bar.** No worksheet today — just a blank diagram box. Tell them: *"Same questions, new tool, no scaffolding. You already know what to ask."*
- **(10 min) Pick a new workflow.** Each student chooses a *different* tool or workflow than Session 1 — the LMS, a shared drive, a photo backup, a group doc, a messaging app used for school.
- **(20 min) Draw it from scratch.** On the blank template, students diagram: where the data starts, where it travels, where it ends up, which account owns it, and who can see it. The only rule: it must answer "where does it live, and who can see it?" Circulate and ask one student per table to defend an arrow.
- **(10 min) Pair-check.** Pairs trade diagrams and try to find one place data could be exposed or lost. They mark it. Collect the diagrams.

Session 3 — Self-directed audit + the change (45 min) — ***fully released***

- **(5 min) Hand it over completely.** No worksheet, no chosen tool, no prompts. *"You pick the tool. You design the diagram. You write the questions worth asking about it."*
- **(15 min) AI focus: what does it do with my words?** Each student diagrams, in their own design, what an AI tool does with the text they type into it: my typed text → where it goes → what comes back → who/what keeps a copy. They don't need technical depth — they need a *mental model* honest enough to predict behavior.
- **(10 min) Write your own questions.** Students write three questions a careful person should ask before trusting this tool with something private.
- **(15 min) The close — one thing you'll do differently.** Go around (or written exit ticket): **one thing you will now do differently, and why.** Examples land like "I'm moving everything to my school account so it doesn't vanish in June,"

or "I won't paste my personal stuff into that AI." Collect these as the capstone artifact.

Student handout (copy-ready)

Where Does It Live?

Software is not magic. Everything you make lives *somewhere*, and someone — or something — can see it. Your job is to find out where, and who.

Part 1 — Audit three projects. Fill this in for three of your own recent school projects:

	Project 1	Project 2	Project 3
What is the file called?			
What folder is it in?			
On this device, in the cloud, or both?			
Which account owns it? (school / personal / the app's own)			
Could you find and open it in 30 seconds right now? (Y/N)			

Part 2 — Trace the "submit." Pick one assignment you turned in. Draw the path it took. Fill in every box you can; put a **?** where you don't know yet:

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My device → [ ___ ] → [ ___ ] → where it lands → who can open it
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Part 3 — Draw the diagram (blank template). For a *new* tool or workflow, draw — on your own — where the data starts, where it travels, where it ends up, which account owns it, and who can see it. The only rule: your diagram must answer **"where does it live, and who can see it?"**

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+------------------------------------------------+
|                                                |
|   (draw it here — boxes, arrows, accounts)     |
|                                                |
+------------------------------------------------+
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Part 4 — The AI question. Diagram what an AI tool does with the text you type into it: *my words → where they go → what comes back → who or what keeps a copy.* Then write **three questions** a careful person should ask before trusting it with something private.

The close. Name **one thing you will now do differently — and why.**

A good audit answer is specific ("it's in my *school* Google Drive, in the 'Bio' folder, and my teacher and I can see it"). **A weak one** is "it's just on my computer somewhere." Aim for the first kind.

What proficient looks like (teacher look-fors)

- The student can **name the actual location** of a file — device or cloud, which folder, **which account** — not just the app that opens it.

- They can **find and retrieve** a specific file fast, without flailing through search.
- Their "submit" trace and AI diagram show a **plausible mental model**, with honest **?** marks where they don't know rather than hand-waving.
- By Session 3 they ask **"where does it live, and who can see it?"** on their own, about a tool nobody told them to examine.

Assessment

Score with `toolkit/rubrics.md`, Digital & Systems Fluency — no separate grade required; lay the language over the diagrams and audit students already produced:

- *"...locate where a file lives and retrieve it immediately."*
- *"...describe a basic mental model of how a piece of software does its job."*
- *"...troubleshoot when technology does not behave as expected."*

Proficient = does it independently this time — finds the file, draws an honest model, asks the right questions without being told. **Advanced** = applies the mental model to *predict* behavior in an unfamiliar tool, or maintains a file/account system organized well enough that someone else could navigate it.

The rubric for this unit

Lay these competency rows over the diagrams and audit students already produced — no separate assignment. **Proficient** is the target (demonstrated and independent); **Advanced** means the student extends the skill unprompted.

Digital & Systems Fluency

The student can…	Not Yet	Approaching	Proficient	Advanced
…locate where a file lives and retrieve it immediately.	Cannot locate files on their own device; does not distinguish local storage, cloud, and downloads folder.	Can find a file eventually but describes its location in vague terms ("it's saved somewhere").	Can state where a file is saved and why (local vs. cloud, which folder, which account) and retrieve it immediately.	Maintains a consistent storage system, can explain it to a peer, and recovers quickly when a sync or access issue occurs.
…describe a basic mental model of how a piece of software does its job.	Treats software as a black box; has no model for what happens between input and output.	Can describe the general purpose of a tool but not how data moves through it.	Gives a simple, accurate model: input goes in, it is processed this way, output comes out — without jargon.	Applies the mental model to predict what will happen in an unfamiliar tool or when something goes wrong.
…troubleshoot when technology does not behave as expected.	Stops working and waits for someone else to fix the problem.	Tries one thing (usually restarting) and escalates immediately if it does not work.	Works through a small set of logical steps, narrows down the likely cause, and escalates with a specific description of what was tried.	Diagnoses systematically, searches for solutions independently, and documents the fix so it can be reused.

Gradual release across the three audits

	Teacher provides	Student provides
Audit 1	Guided worksheet; models the first project; whole-class debrief	Two audited projects + a traced "submit," with support

	Teacher provides	Student provides
Audit 2	A blank diagram box and nothing else	Chooses a new tool; diagrams it independently; pair-checks
Audit 3	Nothing — no tool, no template, no prompts	Picks the tool, designs the diagram, writes the questions, names the change

The goal is that by Audit 3, asking *"where does it live?"* is **automatic, not assigned**.

Differentiation & adaptation

- **Grades 3–5:** Audit one project, not three. Use the concrete frame "find your file and bring it back" — make it a timed treasure hunt. Skip the AI diagram; just ask "where does this live, on the iPad or on the internet?"
- **College / CTE:** Require a written data-flow map for a real tool in their field (the LMS, a code repo, a clinical or POS system), including where credentials and personal data are stored, and one risk they'd flag to an employer.
- **Multilingual learners / IEPs:** Provide the diagram pre-labeled with boxes ("device," "cloud," "account," "who sees it") so the student fills in arrows and contents rather than inventing the whole structure. Sentence stems: "It lives in ___." "It travels to ___." "The people who can see it are ___."
- **Time-crunched version:** Run only Audit 1 on the launch day of a project you were already assigning. Even just mapping where three files live breaks the "software is magic" spell.

Common pitfalls

- **Chasing technical depth.** This is not a networking lesson. If students are debating packet routing, you've overshot. The target is the *habit of asking*, not a correct answer key.
- **Grading the diagram for being "right."** Grade it for being **specific, honest, and self-directed** — the **?** marks are evidence of real thinking, not failure.
- **Doing the auditing for them.** If you fill in the account names, you've taught them nothing. Let them log in and look. Discomfort is the lesson.
- **Running the three audits back-to-back.** The habit forms by *spacing* the audits so the scaffolding can actually withdraw.

The low-lift version (if you have no room for the full unit)

Take any assignment you already give and add a **"plan-before-you-start" memo** worth five points: before doing the work, the student states **where it will live** — file name, folder, and account. That's the weave-in move — the entire unit compressed into one line on a rubric. Five points well spent in the AI era, because a student who can't say where their own file lives can't be trusted to know where their words go when they paste them into something smarter than the app.